

Math 118 – Lecture 04

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1 Lecture 4

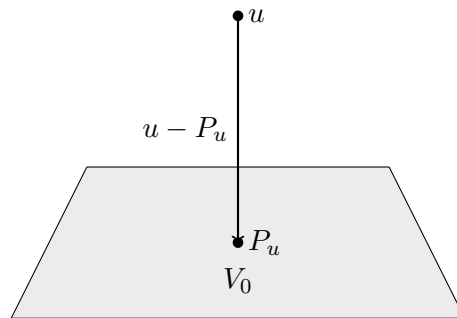
Last time, if $e_1, e_2, \dots, e_n$ is an orthonormal basis for $V_0 \subseteq V$, the orthogonal projection $P : V \rightarrow V_0$ is given by

$$P_u = \sum_{n=1}^N \langle u, e_n \rangle e_n.$$

- P is linear.
- $\langle Pu, v \rangle = \langle u, Pv \rangle$.
- If $v_0 \in V_0$, then $Pv_0 = v_0$.
- If $u \in V$ and $v \in V_0$, then $\langle u - Pu, v \rangle = 0$.

Theorem:

$$\langle u - u_0, v \rangle = 0, \forall v \in V_0 \iff \|u - u_0\| = \min_{v_0 \in V_0} \|u - v_0\|.$$



Geometric Proof of Cauchy: We prove the contrapositive, $\neg A \implies \neg B$.

Suppose $v_0 \in V_0$ and u_0 is the closest point to u in V_0 . Since $u_0, v_0 \in V_0$, we have

$$u_0 - v_0 \in V_0.$$

Set

$$v = u_0 - v_0.$$

Then, by the assumption,

$$\langle u - u_0, v \rangle = 0.$$

Therefore,

$$\langle u - u_0, u_0 - v_0 \rangle = 0.$$

This gives the orthogonality condition in the theorem, and hence

$$\langle u - u_0, v \rangle = 0, \forall v \in V_0 \iff \|u - u_0\| = \min_{v_0 \in V_0} \|u - v_0\|.$$

Notice $V = V_0 \oplus V_0^\perp$

2 Norm of operators

Let X, Y be normed vector spaces. A linear operator $A : X \rightarrow Y$ is bounded if there is a constant M such that is the "operator norm" of A :

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|$$

To show that $\|A\| = M$, you should either:

1: Show that $\|Ax\| \leq M\|x\|, \forall x \in X$

2: Show that either produce an x_0 such that $\|Ax_0\| = M\|x_0\|$ or show that if $K < M$ then $\exists x_0 \neq 0$, such that $\|Ax_0\| > K\|x_0\|$

Definition 2.1 (First definition). Write the definition and one sentence explaining why it matters.

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example). Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?

- TODO: